

Unit 6: Cubic & Cube Root Functions

Mixed Transformations Practice — Advanced

Name: _____

Period: _____

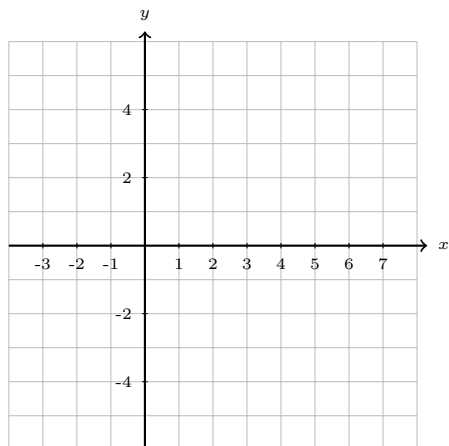
Reminder: $f(x) = a(b(x-h))^3 + k$ or $f(x) = a\sqrt[3]{b(x-h)} + k$ where b controls horizontal stretch/compression

1. $f(x) = 3\left(\frac{1}{2}(x-2)\right)^3 - 1$

$a = \underline{\hspace{1cm}} \quad b = \underline{\hspace{1cm}} \quad h = \underline{\hspace{1cm}} \quad k = \underline{\hspace{1cm}}$

Center: (,)

Parent	New
(,)	(,)
(,)	(,)
(,)	(,)
(,)	(,)
(,)	(,)



x -int: _____

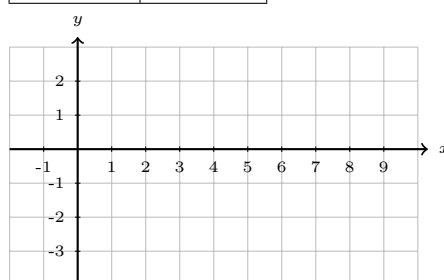
y -int: _____

2. $f(x) = \sqrt[3]{2(x-4)} - 1$

$a = \underline{\hspace{1cm}} \quad b = \underline{\hspace{1cm}} \quad h = \underline{\hspace{1cm}} \quad k = \underline{\hspace{1cm}}$

Center: (,)

Parent	New
(,)	(,)
(,)	(,)
(,)	(,)
(,)	(,)
(,)	(,)



x -int: _____

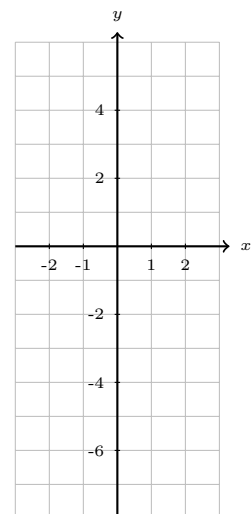
y -int: _____

3. $f(x) = \frac{1}{2}(2x)^3 - 2$

$a = \underline{\hspace{1cm}} \quad b = \underline{\hspace{1cm}} \quad h = \underline{\hspace{1cm}} \quad k = \underline{\hspace{1cm}}$

Center: (,)

Parent	New
(,)	(,)
(,)	(,)
(,)	(,)
(,)	(,)
(,)	(,)



x -int: _____

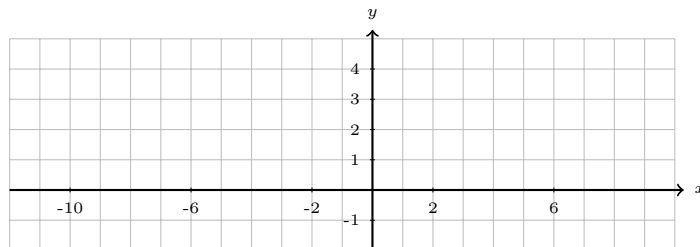
y -int: _____

4. $f(x) = -\sqrt[3]{\frac{1}{2}(x+8)} + 2$

$a = \underline{\hspace{1cm}} \quad b = \underline{\hspace{1cm}} \quad h = \underline{\hspace{1cm}} \quad k = \underline{\hspace{1cm}}$

Center: (,)

Parent	New
(,)	(,)
(,)	(,)
(,)	(,)
(,)	(,)
(,)	(,)



x -int: _____

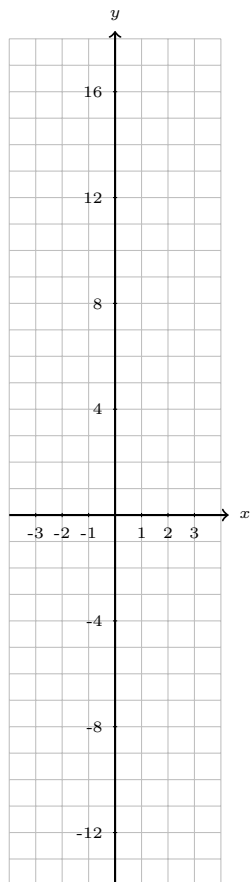
y -int: _____

5. $f(x) = -2(x + 1)^3 + 4$

$a = \underline{\hspace{1cm}}$ $b = \underline{\hspace{1cm}}$ $h = \underline{\hspace{1cm}}$ $k = \underline{\hspace{1cm}}$

Center: ($\hspace{0.5cm}$, $\hspace{0.5cm}$)

Parent	New
($\hspace{0.5cm}$, $\hspace{0.5cm}$)	($\hspace{0.5cm}$, $\hspace{0.5cm}$)
($\hspace{0.5cm}$, $\hspace{0.5cm}$)	($\hspace{0.5cm}$, $\hspace{0.5cm}$)
($\hspace{0.5cm}$, $\hspace{0.5cm}$)	($\hspace{0.5cm}$, $\hspace{0.5cm}$)
($\hspace{0.5cm}$, $\hspace{0.5cm}$)	($\hspace{0.5cm}$, $\hspace{0.5cm}$)
($\hspace{0.5cm}$, $\hspace{0.5cm}$)	($\hspace{0.5cm}$, $\hspace{0.5cm}$)



x -int: $\underline{\hspace{2cm}}$

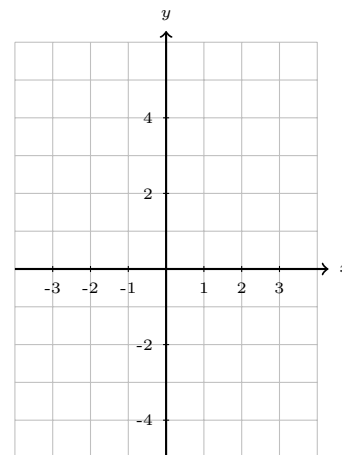
y -int: $\underline{\hspace{2cm}}$

6. $f(x) = 2\sqrt[3]{3x} + 1$

$a = \underline{\hspace{1cm}}$ $b = \underline{\hspace{1cm}}$ $h = \underline{\hspace{1cm}}$ $k = \underline{\hspace{1cm}}$

Center: ($\hspace{0.5cm}$, $\hspace{0.5cm}$)

Parent	New
($\hspace{0.5cm}$, $\hspace{0.5cm}$)	($\hspace{0.5cm}$, $\hspace{0.5cm}$)
($\hspace{0.5cm}$, $\hspace{0.5cm}$)	($\hspace{0.5cm}$, $\hspace{0.5cm}$)
($\hspace{0.5cm}$, $\hspace{0.5cm}$)	($\hspace{0.5cm}$, $\hspace{0.5cm}$)
($\hspace{0.5cm}$, $\hspace{0.5cm}$)	($\hspace{0.5cm}$, $\hspace{0.5cm}$)
($\hspace{0.5cm}$, $\hspace{0.5cm}$)	($\hspace{0.5cm}$, $\hspace{0.5cm}$)

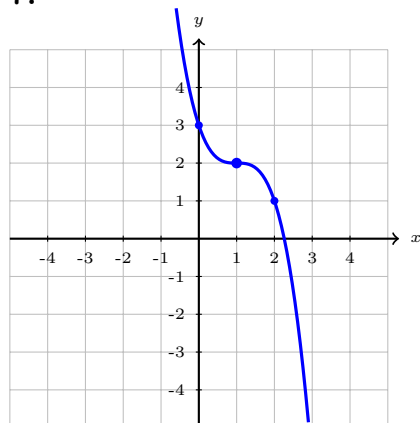


x -int: $\underline{\hspace{2cm}}$

y -int: $\underline{\hspace{2cm}}$

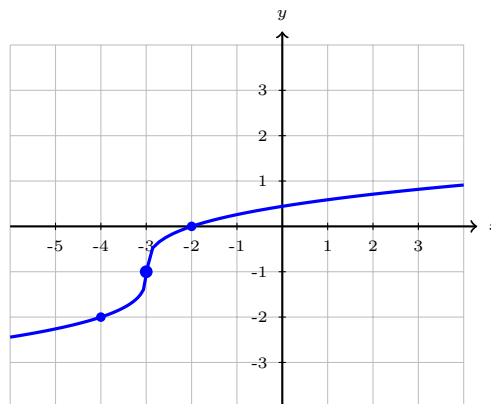
Write the equation from each graph:

7.



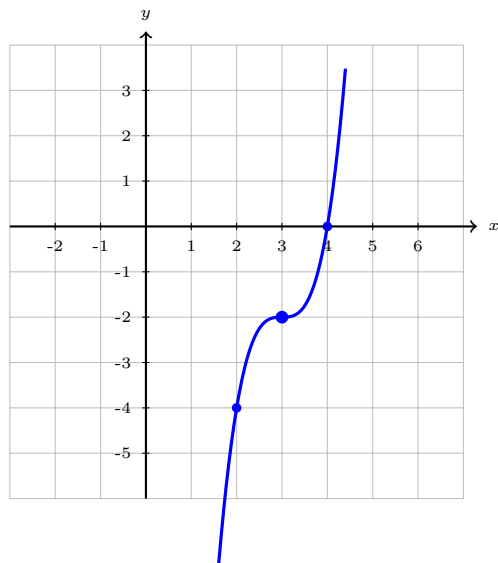
$f(x) = \underline{\hspace{4cm}}$

8.



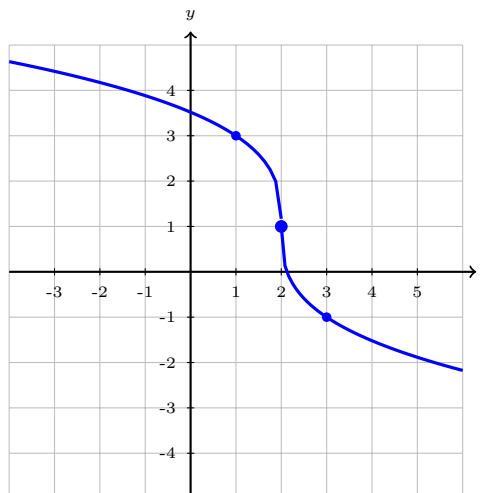
$f(x) = \underline{\hspace{4cm}}$

9.



$$f(x) = \underline{\hspace{2cm}}$$

10.



$$f(x) = \underline{\hspace{2cm}}$$

Write the equation from the description:

11. A cubic function that is vertically stretched by a factor of 2, horizontally compressed by a factor of $\frac{1}{3}$, reflected over the x -axis, shifted left 4, and shifted up 5.

$$f(x) = \underline{\hspace{2cm}}$$

12. A cube root function that is vertically compressed by a factor of $\frac{1}{2}$, horizontally stretched by a factor of 2, shifted right 6, and shifted down 3.

$$f(x) = \underline{\hspace{2cm}}$$

Error Analysis:

13. Marcus says the function $f(x) = \sqrt[3]{2(x-3)} + 1$ has a horizontal stretch by a factor of 2. Explain his error and give the correct transformation.

14. Destiny graphed $f(x) = (x+4)^3 - 2$ with a center point at $(4, -2)$. What mistake did she make?

Compare and Contrast:

15. Without graphing, explain how $f(x) = 2x^3$ and $g(x) = (2x)^3$ are different. Which one is steeper? Justify your answer.

16. The functions $f(x) = 8x^3$ and $g(x) = (2x)^3$ produce identical graphs. Explain why this is true algebraically.

Challenge Problems — For students who finish early

17. A cubic function has its center point at $(1, 3)$ and passes through the point $(2, 5)$. Write the equation of the function.

18. A cube root function has its center point at $(-2, 4)$. If $f(6) = 8$, find the value of a and write the equation.

19. A cubic function satisfies $f(0) = 1$ and $f(2) = -15$ with a center point at $(1, 3)$.

(a) Use one of the points to find the value of a .

(b) Verify your answer using the other point. Show your work.

(c) Write the final equation: $f(x) =$ _____